

Security+ 3.3 · Data Types & Classifications

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.3 — Data Types & Classifications (5:54)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Classify data: PUBLIC (no harm if disclosed); INTERNAL (internal use only); CONFIDENTIAL (strategy, PII — leakage = severe damage); RESTRICTED (trade secret — leakage = catastrophic). Classification DETERMINES the control. PHI = health data; PCI-DSS = card data.

High-Yield Core Concepts Tested on Exam

- Classification determines WHICH protection each data set receives
- PHI = health; PCI-DSS = card; PII = personal data

Exam Tips & Classic Scenarios

- 'least restrictive classification' = public; 'most restrictive' = restricted

Typical Exam Scenario: 'customer data with name, address, SSN' = PII — confidential classification.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.3 — Data Types & Classifications)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **DLP:** Data Loss Prevention: Software/hardware safeguards monitoring, identifying, and preventing unauthorized exfiltration of sensitive organizational assets in use, transit, or rest.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. In the SHARED RESPONSIBILITY model, who handles the PHYSICAL security of the datacenter?

- A)** The customer
- B)** The cloud provider
- C)** Both equally
- D)** An external auditor

2. A control that prevents an employee from sending a confidential spreadsheet outside the company is:

- A) DLP (Data Loss Prevention)**
- B) SSO**
- C) MFA**
- D) VPN**

3. Corporate scenario (Data Types & Classifications): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.3), an analyst evaluates the following technical requirement: 'Classification determines WHICH protection each data receives'. What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Replace preventive technical controls with informal terms of adherence**
- C) Allow shared access among administrators with generic credentials**
- D) Implement the corresponding solution: Classification determines WHICH protection each data receives...**

4. Corporate scenario (Data Types & Classifications): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.3), an analyst evaluates the following technical requirement: 'PHI = health; PCI-DSS = card; PII = personal data'. What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Replace preventive technical controls with informal terms of adherence**
- C) Allow shared access among administrators with generic credentials**
- D) Implement the corresponding solution: PHI = health; PCI-DSS = card; PII = personal data...**

5. Corporate scenario (Data Types & Classifications): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.3), an analyst evaluates the following technical requirement: 'least restrictive classification'=public; 'most restrictive'=restricted'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to speed up system throughput
- B)** Implement the corresponding solution: 'least restrictive classification'=public; 'most restrictive'=restricted...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

6. Corporate scenario (Data Types & Classifications): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.3), an analyst evaluates the following technical requirement: 'customer data with name, address, SSN' = PII — confidential classification. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'customer data with name, address, SSN' = PII — confidential classification...

7. Corporate scenario (Data Types & Classifications): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.3), an analyst evaluates the following technical requirement: 'Classification determines WHICH protection each data receives.' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Classification determines WHICH protection each data receives...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Data Types & Classifications): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.3), an analyst evaluates the following technical requirement: 'PHI = health; PCI-DSS = card; PII = personal data'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: PHI = health; PCI-DSS = card; PII = personal data...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

9. Corporate scenario (Data Types & Classifications): During a security review and compliance audit aligned with the CompTIA Security+ (SY0-701, Section 3.3) exam, an analyst evaluates the following technical requirement: "'least restrictive classification'=public; 'most restrictive'=restricted'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'least restrictive classification'=public; 'most restrictive'=restricted...
- D)** Allow shared access among administrators using generic credentials

10. Corporate scenario (Data Types & Classifications): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.3), an analyst evaluates the following technical requirement: 'customer data with name, address, SSN'=PII — confidential classification. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'customer data with name, address, SSN'=PII — confidential classification...
- D)** Allow shared access among administrators using generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — The cloud provider.

✓ **Why it is correct:** In the cloud computing shared responsibility model, the physical security of the datacenter — including facility access control, video surveillance, redundant power, climate control, fire prevention and suppression, and protection against natural disasters — is the exclusive responsibility of the cloud provider. CompTIA Security+ SY0-701 emphasizes that the provider is always responsible for 'security OF the cloud' (the underlying infrastructure), while the customer is responsible for 'security IN the cloud' (their data, configurations, identities, and applications). The customer has no physical access to the datacenter and therefore cannot implement physical controls; the responsibility is non-transferable and not shared in this specific aspect.

✗ **Why the others are wrong:**

- *The customer:* The customer is responsible for security IN the cloud, i.e., their data, access configurations, encryption of their information, identity management, and patching of their instances. However, they do not have physical access to the provider's datacenter and cannot install locks, cameras, or fire suppression systems. Assigning physical security to the customer would violate the shared responsibility model, as they do not control the physical environment. The question specifically asks about the PHYSICAL security of the datacenter, which is the provider's exclusive domain.
- *Both equally:* Shared responsibility is not divided equally in all aspects. It is segmented by layers: the provider handles physical security, virtualization infrastructure, and the underlying network; the customer handles the security of their data, applications, and identities. The physical security of the datacenter is a layer the customer cannot operate or directly audit, so there is no equal sharing. Saying both are equally responsible for physical security is technically incorrect and contradicts the model defined by CompTIA and major cloud providers (AWS, Azure, GCP).
- *An external auditor:* An external auditor performs independent compliance assessments, such as SOC 2, ISO 27001, or PCI DSS certifications, but does not execute the physical security of the datacenter. They verify whether the controls implemented by the provider are adequate, but they are not responsible for operating them. Auditing is a verification function, not an operational one. Therefore, assigning physical security to an external auditor confuses the oversight role with the execution role, which is a serious conceptual error in the context of the shared responsibility model.

2) A — DLP (Data Loss Prevention).

✓ **Why it's correct:** DLP (Data Loss Prevention) is a set of technologies and policies aimed at preventing the exfiltration of confidential data by monitoring and controlling transfers of sensitive information. In CompTIA Security+ (SY0-701), DLP is presented as a control that can block the sending of data outside the organization, whether via email, web, USB, or other channels. The question describes exactly a control that prevents an employee from sending a confidential spreadsheet outside the company, which is the primary function of DLP.

✗ **Why the others are wrong:**

- *SSO:* SSO (Single Sign-On) is an authentication mechanism that allows a user to access multiple systems with a single credential. It does not control data transfer or prevent information exfiltration. Therefore, it does not apply to the described scenario.
- *MFA:* MFA (Multi-Factor Authentication) is an authentication method that requires two or more verification factors to grant access. It strengthens authentication but does not prevent an authorized user from sending confidential data outside. It is not a data loss prevention control. Thus, it is not the correct answer.

- **VPN:** VPN (Virtual Private Network) is a technology that creates an encrypted tunnel for secure communication, but it does not have the function of inspecting or blocking the transfer of confidential data. It protects transmission but does not prevent exfiltration by an authorized user. Therefore, it does not meet the requirement of the question.

3) D — Implement the corresponding solution: Classification determines WHICH protection each data receives....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.3 (Data Types & Classifications): Classification determines WHICH protection each data receives.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

4) D — Implement the corresponding solution: PHI = health; PCI-DSS = card; PII = personal data....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.3 (Data Types & Classifications): PHI = health; PCI-DSS = card; PII = personal data. PHI (Protected Health Information) refers to health-related data protected under HIPAA. PCI-DSS (Payment Card Industry Data Security Standard) governs cardholder data. PII (Personally Identifiable Information) includes personal data that can identify an individual. Proper classification ensures appropriate handling, storage, and protection controls are applied based on data sensitivity and regulatory requirements.

✗ **Why the other options are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

5) B — Implement the corresponding solution: 'least restrictive classification'=public; 'most restrictive'=restricted....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.3 (Data Types & Classifications): 'least restrictive classification'=public; 'most restrictive'=restricted.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to speed up system throughput:* Directly violates the Accounting

pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).

- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) D — Implement the corresponding solution: 'customer data with name, address, SSN' = PII — confidential classification....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.3 (Data Types & Classifications): 'customer data with name, address, SSN' = PII — confidential classification. Personally Identifiable Information (PII) includes data that can be used to identify an individual, such as name, address, and Social Security Number. It should be classified as confidential and protected accordingly.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) A — Implement the corresponding solution: Classification determines WHICH protection each data receives....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.3 (Data Types & Classifications): Classification determines WHICH protection each data receives. Data classification (e.g., public, private, confidential, restricted) dictates the level of security controls applied, ensuring that sensitive data receives appropriate protection based on its value and regulatory requirements.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) B — Implement the corresponding solution: PHI = health; PCI-DSS = card; PII = personal data....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.3 (Data Types & Classifications): PHI = health; PCI-DSS = card; PII = personal data.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
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- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

9) C — Implement the corresponding solution: 'least restrictive classification'=public; 'most restrictive'=restricted....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.3 (Data Types & Classifications): 'least restrictive classification'=public; 'most restrictive'=restricted. Data classification labels determine handling requirements, with public data requiring the least protection and restricted data requiring the highest level of access control and encryption.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) C — Implement the corresponding solution: 'customer data with name, address, SSN'=PII — confidential classification....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.3 (Data Types & Classifications): 'customer data with name, address, SSN'=PII — confidential classification. PII (Personally Identifiable Information) must be classified as confidential and protected with appropriate technical and administrative controls, such as encryption, access control, and data loss prevention.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
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